Mohammad Owais, *Undergraduate Researcher*

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Education

Expected March 2027 — **Abbottabad University of Science and Technology (AUST)**
Abbottabad, Pakistan
Major: BS Computer Science (8th Semester, In Progress)
BS Final Year Project: *Neural Image and Video Super-Resolution Engine* (Supervisor: Ms. Naveeda Bibi)
Relevant Coursework: Artificial Intelligence, Digital Image Processing, Linear Algebra, Probability & Statistics, Data Structures & Algorithms, Design and Analysis of Algorithms, Fundamentals of Data Mining

Research Experience

Brain Tumor MRI Super-Resolution Benchmarking (Without Fine-Tuning) *Jun 2026 – Oct 2026*
First & Corresponding Author — In collaboration with Faculty of Systems Design, Tokyo Metropolitan University, Japan

- Benchmarked five 4× SR methods (bicubic, EDSR, Waifu2x, Real-ESRGAN, SwinIR) with frozen pretrained weights across 2,000 public brain tumor MRI slices (four diagnostic classes).
- Designed a reproducible evaluation protocol using shared lossless LR/HR inputs, head-mask PSNR/SSIM, full-frame LPIPS, 64-bit DCT pHash duplicate screening, and Bonferroni-corrected paired Wilcoxon tests.
- Demonstrated that SwinIR achieved the highest fidelity (+3.22 dB over bicubic), whereas Real-ESRGAN lost fidelity (−4.30 dB) and fabricated boundary texture (isolated via a bicubic-trained ESRGAN control); downstream 5-split ResNet-18 classification accuracy rose from 83.95% (bicubic) to 89.60% (SwinIR).

Manuscripts

- **Owais, Mohammad**, & Owais, Muhammad. (2026). "Benchmarking Off-the-Shelf Deep Learning Super-Resolution Models on Public Brain Tumor MRI: A CNN-versus-Transformer Comparison Without Fine-Tuning." *Manuscript under review at Visual Computing for Industry, Biomedicine, and Art* (Springer Nature, SCIE; Submission ID: `VICO-D-26-00680`).

Code and Data

- **GitHub Repository (`Brain-MRI-SR-Benchmark`, MIT License):** Complete end-to-end evaluation pipeline (`SR_Master_Pipeline.ipynb`), 2,000-slice dataset manifest (`manifest.json`), and 10,000-row per-image metrics (`github.com/owaiskhan2501000/Brain-MRI-SR-Benchmark`).
- **Zenodo Archival Record (`v1.0.1`):** Permanently archived benchmark code, statistical summary tables, residual heat-map figures, and 5-seed ResNet-18 classification outputs (`doi:10.5281/zenodo.23467749`).

Research Interests

Medical image analysis and brain MRI super-resolution: perception–distortion trade-off, hallucination and structural fidelity, downstream diagnostic utility, memory-efficient 3D state space architectures (Mamba/SSMs), and anatomy-aware reconstruction using pre-trained medical encoders.

Technical Skills

Programming	Python, C++, SQL.
Deep Learning & Computer Vision	PyTorch, OpenCV, Torchvision, timm; experience with EDSR, Waifu2x, Real-ESRGAN, ESRGAN, SwinIR, and ResNet-18.
Medical Image Processing	NiBabel, SimpleITK, scikit-image; NIfTI volume handling, canonical RAS orientation standardization, N4 bias-field correction, foreground head-mask segmentation, and 3D patch extraction.
Tools & Reproducibility	Git, GitHub, Zenodo, LaTeX, Google Colab (CUDA/T4), FFmpeg, VS Code.
Additional Development	LangChain, FAISS, Whisper, Llama 3.1, JavaScript, TypeScript, React, Next.js.

Technical Projects

Aug 2026 – Sep 2026 — **Brain MRI Preprocessing Pipeline** *Python, PyTorch, NiBabel, SimpleITK*
Developed preprocessing workflows for volumetric MRI data in NIfTI format, including canonical RAS reorientation, N4 bias-field correction, and $64 \times 64 \times 64$ voxel patch extraction.

May 2026 – Present — **Neural Image and Video Super-Resolution Engine (BS Final Year Project)**
Python, PyTorch, Real-ESRGAN, FFmpeg, CUDA
Developed an image and video upscaling pipeline using Real-ESRGAN for neural image reconstruction (*Supervisor: Ms. Naveeda Bibi*).
Implemented spatial tiling and batching to manage GPU memory during inference.
Integrated FFmpeg-based video processing and encoding to maintain a constant output frame rate.

Apr 2026 – May 2026 — **Multimodal Retrieval-Augmented Generation Pipeline** *Python, LangChain, FAISS, Whisper, Llama 3.1*
Built an audio/video transcription and retrieval pipeline using Whisper and FAISS, with Llama 3.1 for summarization and question answering over indexed content.

Selected Certifications & Courses

2026 — **Deep Learning Specialization**
DeepLearning.AI, Coursera

Generative Adversarial Networks (GANs) Specialization
DeepLearning.AI, Coursera

2025 — **Introduction to Computer Vision and Image Processing**
IBM, Coursera

Selected Certifications & Courses (continued)

2024 — **Python for Data Science, AI & Development**
IBM, Coursera

Languages and English Proficiency

English — **Professional proficiency (Medium of Instruction: English).**
University-issued English Proficiency Certificate from Abbottabad University of Science and Technology, Pakistan.

Urdu — Native.

Japanese — Beginner (*studying Hiragana, Katakana, and basic JLPT N5 expressions*).

Academic References

- **Ms. Naveeda Bibi** — *BS Final Year Project Supervisor*
  Assistant Professor, Department of Computer Science
  Abbottabad University of Science and Technology
  Email: naveedabibi@aust.edu.pk
- **Dr. Gohar Zaman** — *Department Coordinator & Course Instructor (Data Mining)*
  Assistant Professor, Department of Computer Science
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